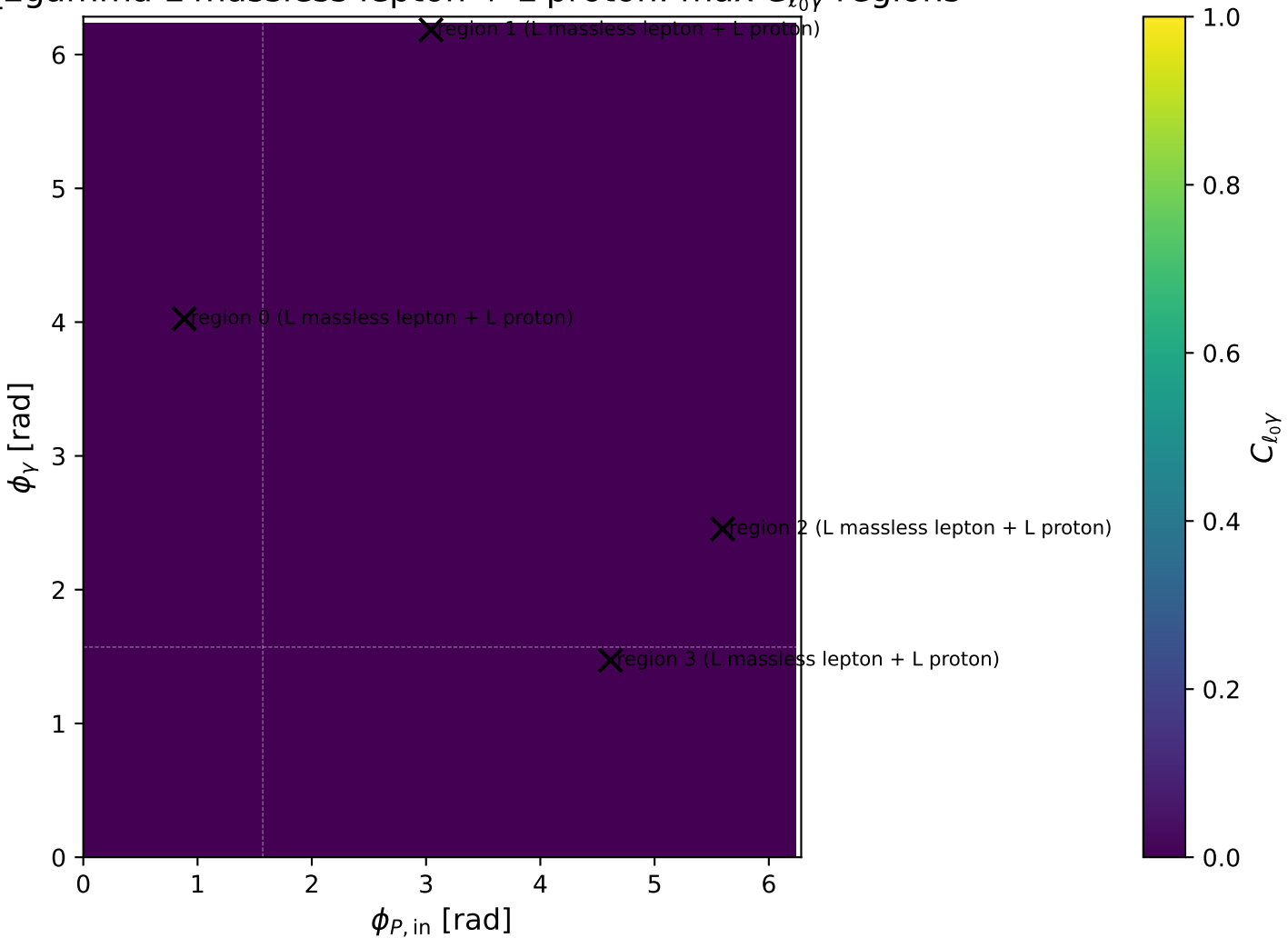
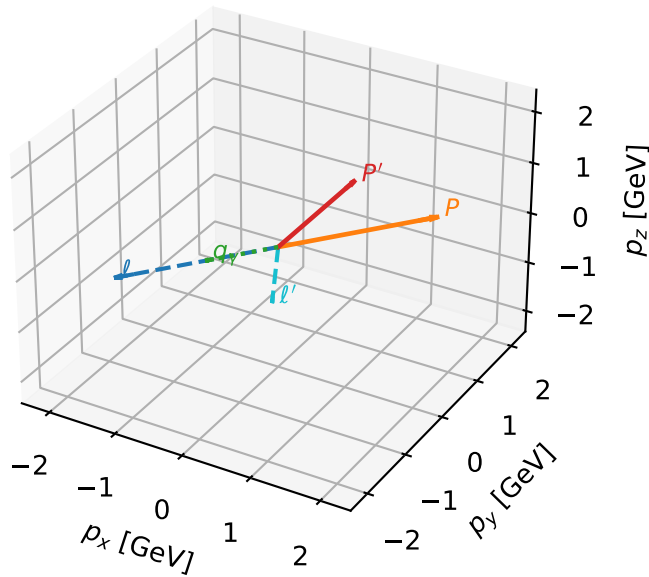


Gamma L massless lepton + L proton: max $C_{\ell_0\gamma}$ regions



L massless lepton + L proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta

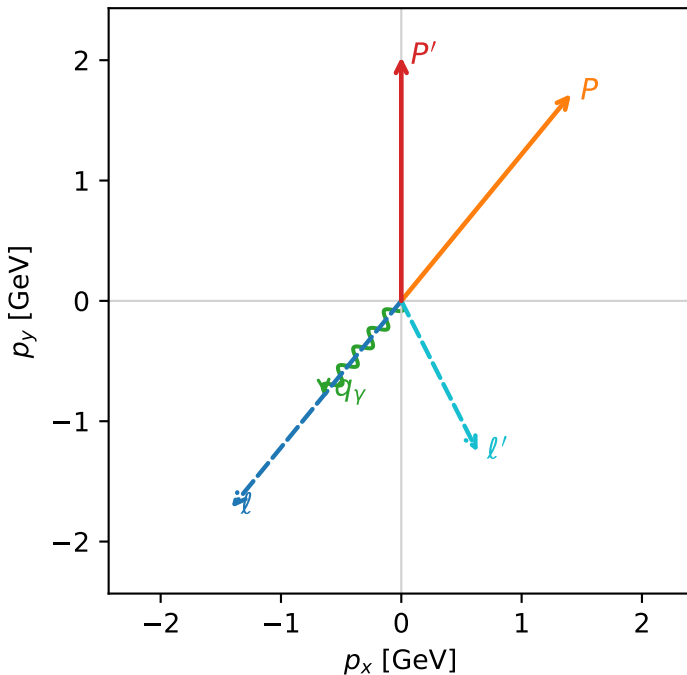


c_e_gamma_mid_egamma_ll_region_0 C_{\ell_0\gamma}=9.71682e-11
 region: mid_Egamma_LL_region_0 final-pair delta_xy=1.15564 rad
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

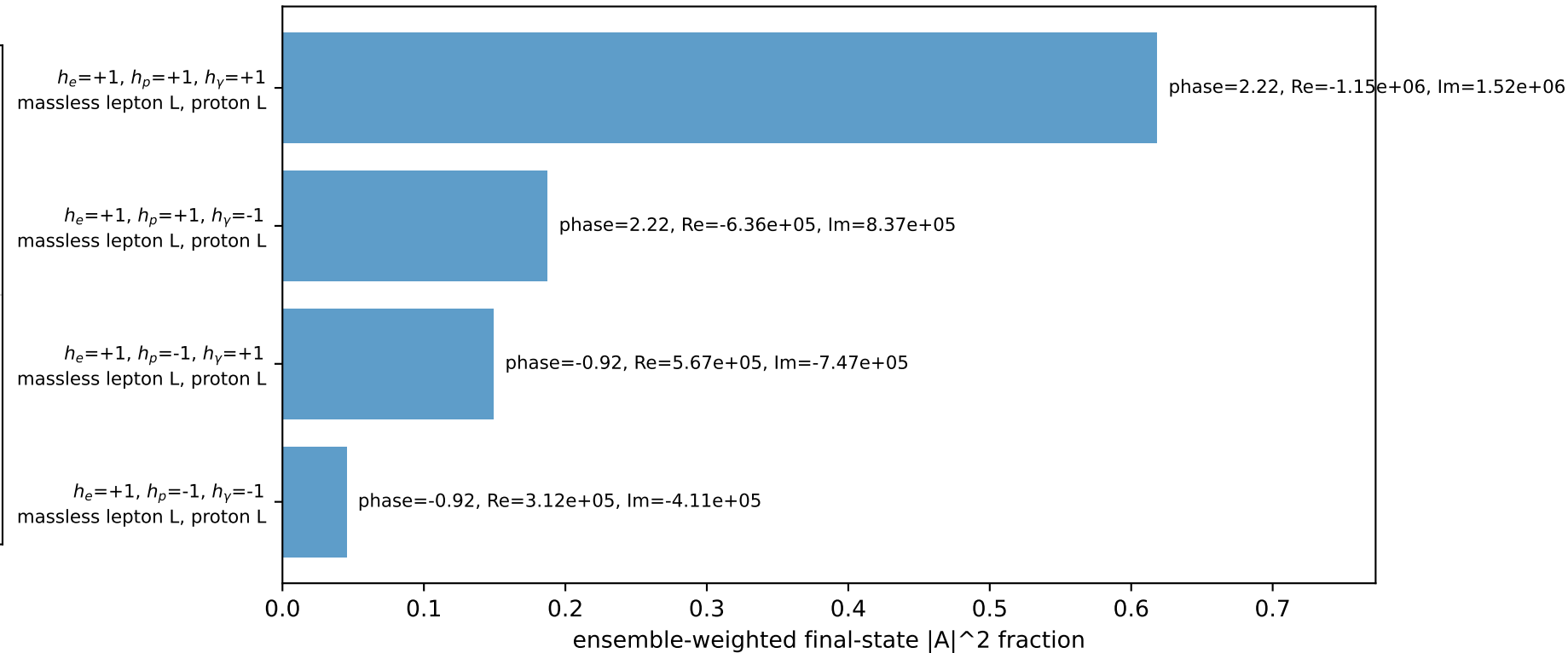
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.02682$
 $\theta_{in}=1.57079$, $\phi_l=4.02517$, $\phi_p=0.883573$
 $E_\gamma=1$, $\phi_\gamma=4.02517$
 $Q^2=3.72996$, $x_B=0.329206$, $t=-2.05314$, $W^2=8.48005$, $y=0.549049$
 $\theta(\ell', \gamma)=66.2131$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -1.4112$ $p_y= -1.7195$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 1.4112$ $p_y= 1.7195$ $p_z= 0.0000$
 ℓ' $E= 1.4052$ $p_x= 0.6344$ $p_y= -1.2538$ $p_z= 0.0000$
 P' $E= 2.2333$ $p_x= 0.0000$ $p_y= 2.0268$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.6344$ $p_y= -0.7730$ $p_z= 0.0000$

Transverse plane

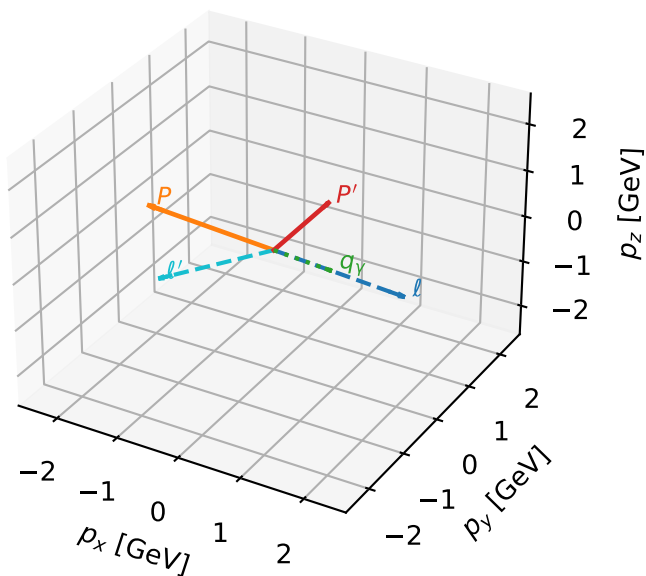


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



L massless lepton + L proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta



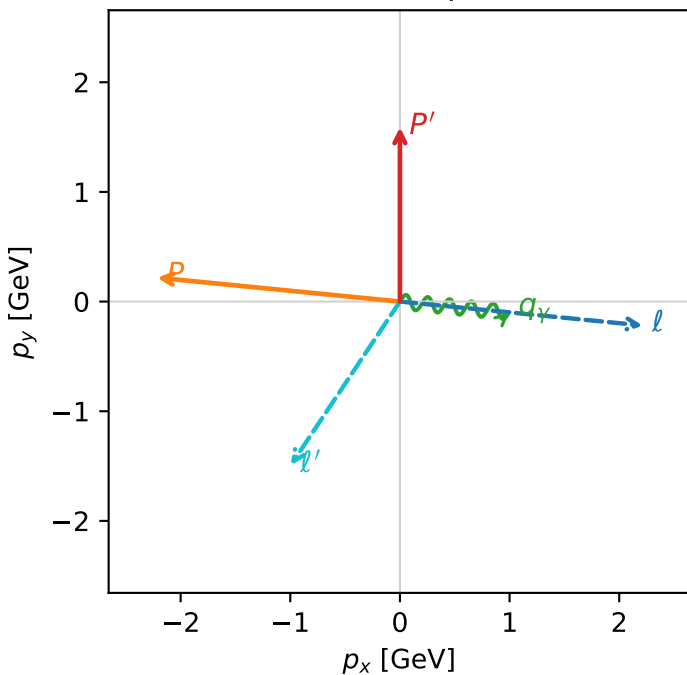
c_e_gamma_mid_egamma_ll_region_1 $C_{\ell_0\gamma}=5.67121e-11$
 region: mid_Egamma_LL_region_1 final-pair delta_xy=2.06105 rad
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.58942$
 $\theta_{in}=1.57079$, $\phi_{\ell}=6.18501$, $\phi_P=3.04342$
 $E_Y=1$, $\phi_Y=6.18501$
 $Q^2=11.7323$, $x_B=0.745618$, $t=-6.45798$, $W^2=4.88255$, $y=0.762503$
 $\theta(\ell', \gamma)=118.089$ deg

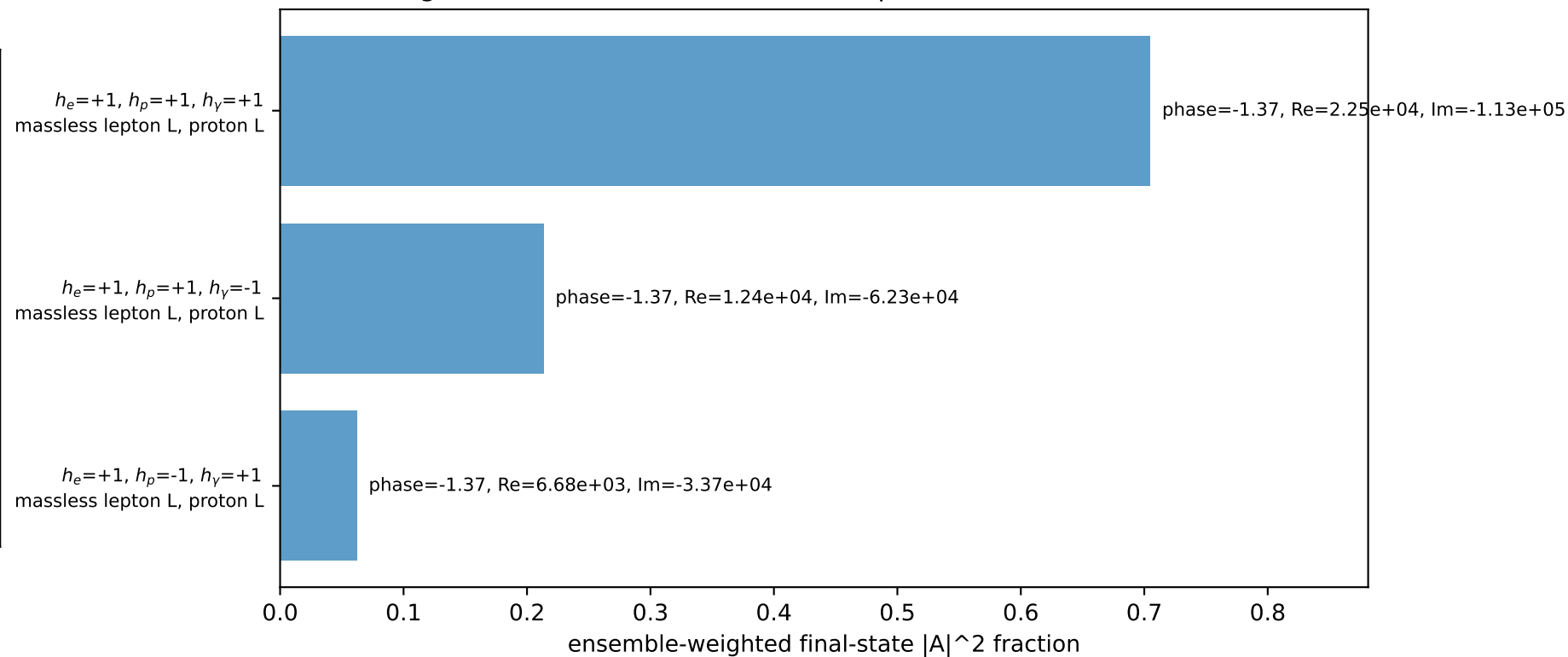
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ	$E=$	2.2244	$p_x=$	2.2137	$p_y=$	-0.2180	$p_z=$	-0.0000
P	$E=$	2.4141	$p_x=$	-2.2137	$p_y=$	0.2180	$p_z=$	0.0000
ℓ'	$E=$	1.7930	$p_x=$	-0.9952	$p_y=$	-1.4914	$p_z=$	0.0000
P'	$E=$	1.8456	$p_x=$	0.0000	$p_y=$	1.5894	$p_z=$	0.0000
q_Y	$E=$	1.0000	$p_x=$	0.9952	$p_y=$	-0.0980	$p_z=$	0.0000

Transverse plane

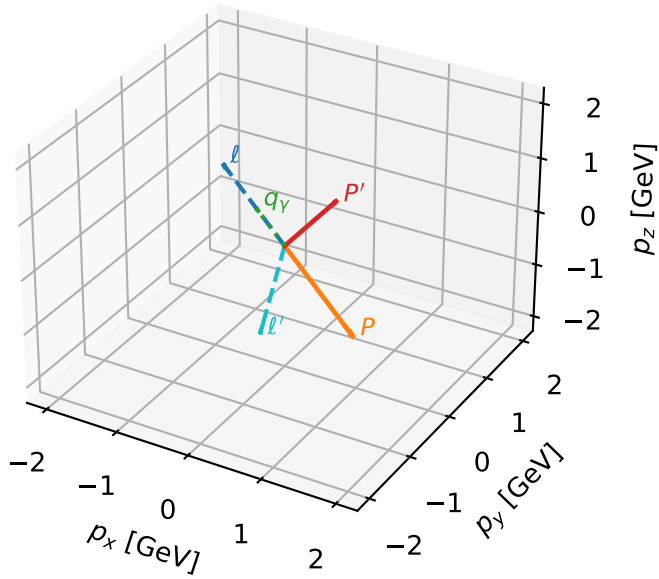


Leading initial-ensemble/final-state components (N=3, retained=98.1%)



L massless lepton + L proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta

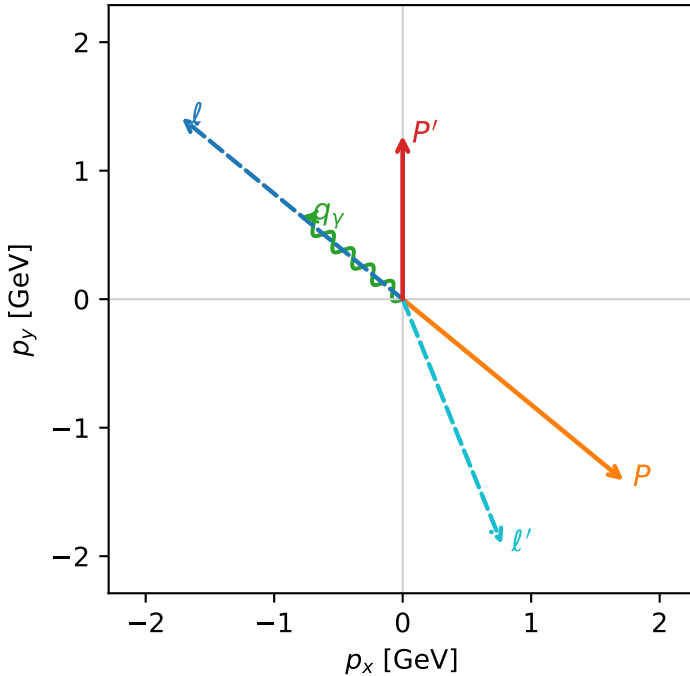


c_e_gamma_mid_egamma_ll_region_2 $C_{\ell_0\gamma}=4.10143\text{e-}11$
 region: mid_Egamma_LL_region_2 final-pair delta_xy=2.64315 rad
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.27253$
 $\theta_{in}=1.57079$, $\phi_l=2.45437$, $\phi_p=5.59596$
 $E_\gamma=1$, $\phi_\gamma=2.45437$
 $Q^2=17.1944$, $x_B=0.917446$, $t=-9.46456$, $W^2=2.42703$, $y=0.908199$
 $\theta(\ell', \gamma)=151.441$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -1.7195$ $p_y= 1.4112$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 1.7195$ $p_y= -1.4112$ $p_z= 0.0000$
 ℓ' $E= 2.0576$ $p_x= 0.7730$ $p_y= -1.9069$ $p_z= 0.0000$
 P' $E= 1.5809$ $p_x= 0.0000$ $p_y= 1.2725$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

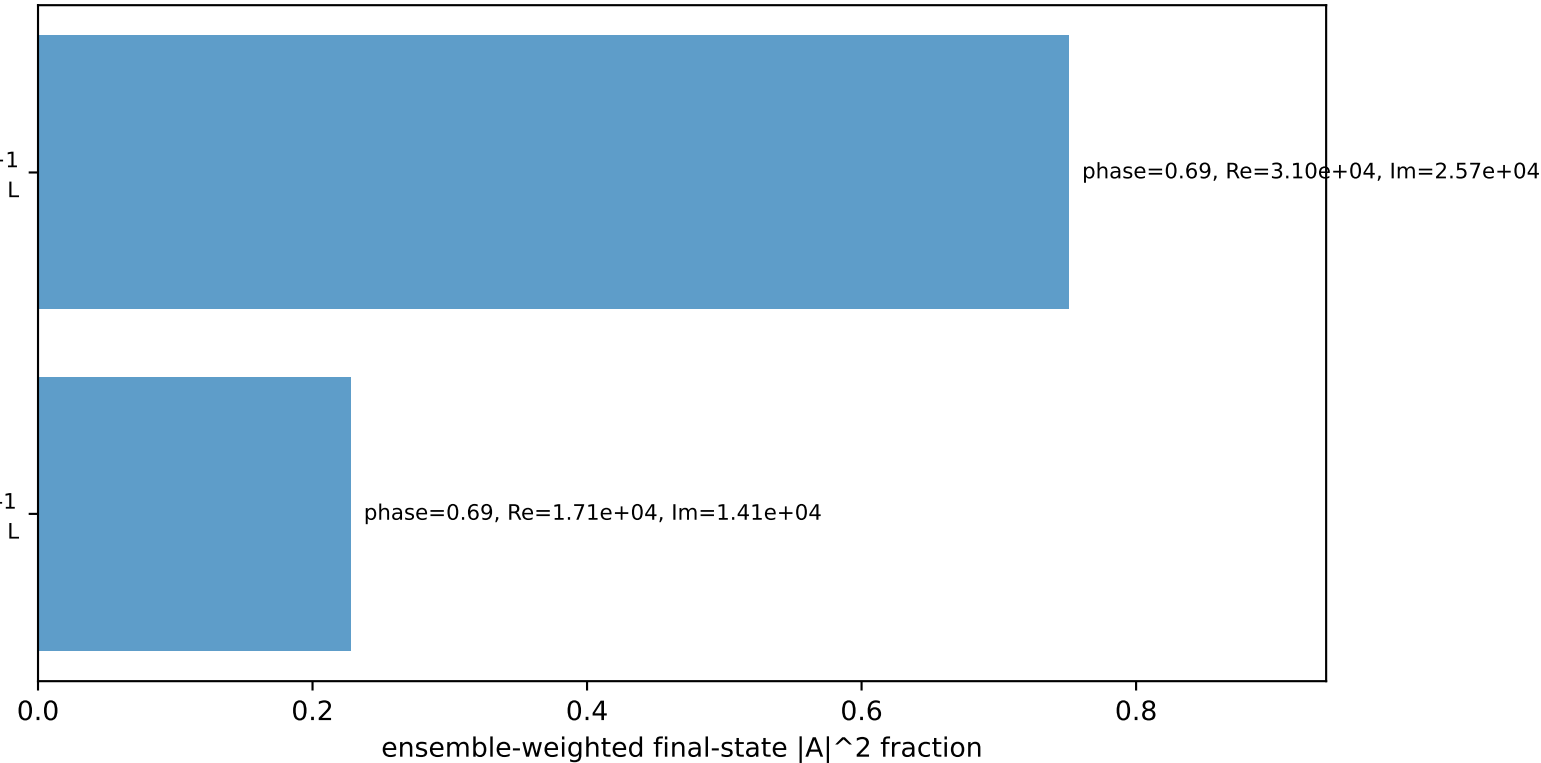
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
 massless lepton L, proton L

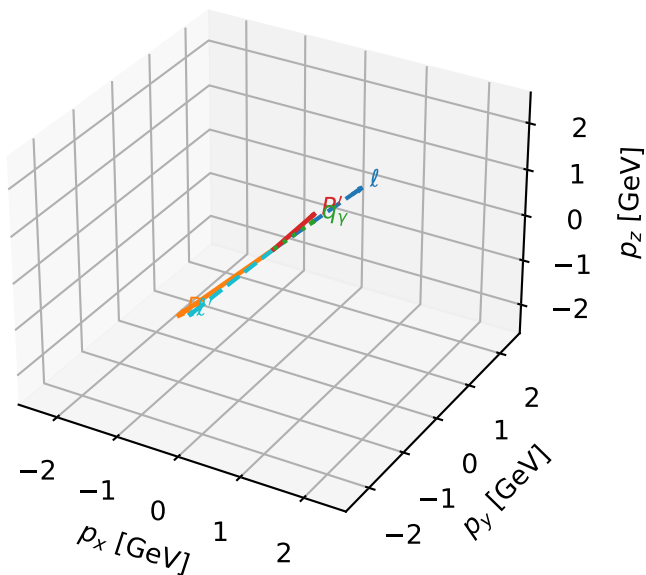
$h_e=+1, h_p=+1, h_\gamma=-1$
 massless lepton L, proton L

Leading initial-ensemble/final-state components (N=2, retained=97.8%)



L massless lepton + L proton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta

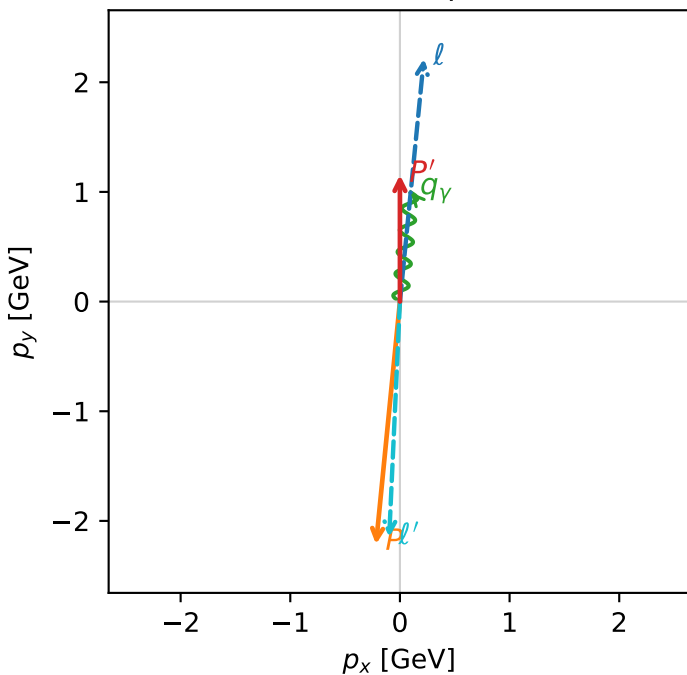


c_e_gamma_mid_egamma_ll_region_3 $C_{\ell_0\gamma}=3.30899\text{e-}11$
 region: mid_Egamma_LL_region_3 final-pair delta_xy=3.08899 rad
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_{\ell_0}\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=1.15398$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.47262$, $\phi_P=4.61421$
 $E_{\gamma}=1$, $\phi_{\gamma}=1.47262$
 $Q^2=19.1292$, $x_B=0.9658$, $t=-10.5296$, $W^2=1.55723$, $y=0.959808$
 $\theta(\ell', \gamma)=176.986$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.2180$ $p_y= 2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.2180$ $p_y= -2.2137$ $p_z= 0.0000$
 ℓ' $E= 2.1514$ $p_x= -0.0980$ $p_y= -2.1492$ $p_z= 0.0000$
 P' $E= 1.4871$ $p_x= 0.0000$ $p_y= 1.1540$ $p_z= 0.0000$
 q_{γ} $E= 1.0000$ $p_x= 0.0980$ $p_y= 0.9952$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=100.0%)

